

TOPIC 15/60

How do LLMs predict tokens?

It's not "thinking."

It's the world's most advanced probability
machine.

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THE INTUITION

An LLM doesn't "know" facts. It knows the mathematical likelihood of what words usually follow each other.

Prompt: "The sky is..."

- Blue (98%)
- Gray (1.5%)
- Red (0.5%)



STEP 1: INPUT



Tokenization

First, the AI slices your text into tiny numerical IDs (Tokens).

"Hello" → [154961

"World" → [2159]

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STEP 2

Embeddings

Mapping Meaning

It converts those IDs into high-dimensional vectors. It places "Hello" and "Hi" near each other in a massive conceptual space.

Mathematical Coordinates

"Hello" = [0.12, -0.45, 0.88, ...]

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STEP 3

Processing

Deep Attention Layers

The vector travels through dozens of Transformer layers, where "Self-Attention" mixes the contexts.

It refines the meaning of each word based on its neighbours. "Apple" starts as a vague concept and becomes "Apple the fruit" because the layer saw the word "orchard" nearby.

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STEP 4

Logits

The Raw Score

At the very last layer, the AI outputs a giant list of scores, one for every single token in its dictionary (e.g., 100,000 possibilities).

Scores:

"cat": 12.5

"dog": 4.2

"airplane": -5.1

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STEP 5: SOFTMAX

NORMALIZATION

Softmax takes those raw scores (Logits) and forces them to add up to 100%.

"cat" becomes 85% probability.

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STEP 6

Sampling

It doesn't always pick the most likely word. If it did, it would be boring and repetitive. It picks based on the "Temperature" setting.

Low Temperature = Robotically logical (picks the 99% answer).

High Temperature = Creative and unpredictable (might pick the 5% answer).

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THE AUTOREGRESSIVE LOOP

The Loop

Once it chooses a token, that token is added to the prompt. The entire process repeats for the next word.

Infinite Generation

This continues until it generates a "STOP" token or hits its limit.

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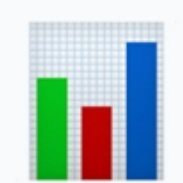


Prediction Loop

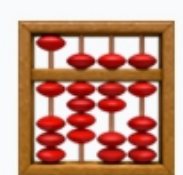
CONCEPT UNLOCKED



Tokenize



Embedding



Logits & Softmax



Autoregressive Loop

CONNECT WITH ME



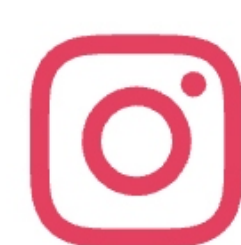
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